ECO5050 AI Applications and Techniques in Economics

Lecture 2: Model Selection - Part 1

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Outline

Model Selection

- In practice, we are faced with the problem of not only estimating parameters given a model but also choosing the model to be estimated.
  - Lag length selection for dynamic models
  - Choosing which variables to include in the model
  - Choosing the functional form of the model
- Model selection methods attempts to choose a single "best" model, where "best" typically depends on the forecaster's loss function, information set, forecast horizon and the underlying data-generating process.
  - In-sample fit: the estimated risk using the full data sample
  - Out-of-sample fit: the estimated risk using a "holdout" sample that was not used for parameter estimation
- If all models are special cases of a "super model" that nests everything, the distinction between model selection and parameter estimation becomes somewhat blurred.

In-sample vs. Out-of-sample risk

- Model selection is related to evaluation of forecast models (which we will learn more later in the course)
- Consider the simple forecasting problem: we want to forecast y_{T+1} under MSE loss given $Z_T = \{y_t\}_{t=1}^T$, where $y_t \overset{i.i.d.}{\sim} (\mu, \sigma^2)$.
- The optimal forecast is μ which is unknown, so let $f(Z_T) = \bar{y}_T = T^{-1} \sum_{t=1}^T y_t$.
- For any in-sample observation $t = 1, \ldots, T$, the MSE is

$$
\begin{aligned}
E[(y_t - \bar{y}_T)^2] &= E\left[\left((y_t - \mu) - T^{-1}\sum_{s=1}^{T}(y_s - \mu)\right)^2\right] \\
&= \sigma^2\left(1 + \frac{T}{T^2} - 2\frac{1}{T}\right) \\
&= \sigma^2(1 - T^{-1})
\end{aligned}
$$

- The third term in the second line comes from the cross product when computing the squared terms

In-sample vs. Out-of-sample risk

- For an out-of-sample observation $T+1$, the MSE is

$$E[(y_{T+1} - \bar{y}_T)^2] = E\left[\left((y_{T+1} - \mu) - T^{-1}\sum_{t=1}^{T}(y_t - \mu)\right)^2\right]$$
$$= \sigma^2\left(1 + \frac{T}{T^2}\right)$$
$$= \sigma^2(1 + T^{-1})$$

- The estimation error reduces the in-sample MSE but increases the out-of-sample MSE.

In-Sample Fit

- Improvements in model fit could either reflect a model's genuine ability to produce better forecasts or, alternatively, be due to the model's tendency to overfit

- Example: Linear models with MSE loss

$$\begin{aligned} M_1 &: y_{t+1} = x_{1t}'\beta + \varepsilon_{1t+1} \\ M_2 &: y_{t+1} = x_{1t}'\gamma_1 + x_{2t}'\gamma_2 + \varepsilon_{2t+1} \end{aligned}$$

  Plug-in OLS estimates $\hat{\beta}$, $\hat{\gamma} = (\hat{\gamma}_1', \hat{\gamma}_2')'$ it follows that

$$T^{-1} \sum_{t=0}^{T-1} L(f_2(z_t, \hat{\gamma}), y_{t+1}) \leq T^{-1} \sum_{t=0}^{T-1} L(f_1(z_t, \hat{\beta}), y_{t+1}) \tag{1}$$

  from the well-known result that the regression R^2 is weakly increasing as additional variables are added.

In-Sample Fit

- Example: Linear models with MSE loss

$$M_1 : y_{t+1} = x_{1t}'\beta + \varepsilon_{1t+1}$$
$$M_2 : y_{t+1} = x_{1t}'\gamma_1 + x_{2t}'\gamma_2 + \varepsilon_{2t+1}$$

- The inequality (1) holds even though it may be that the population expected loss under M_1 is smaller than that under M_2

$$E\left[L(f_2(z_t, \gamma^*), y_{t+1})\right] > E\left[L(f_1(z_t, \beta^*), y_{t+1})\right]$$

  where γ^* and β^* are the probability limits of $\hat{\gamma}$ and $\hat{\beta}$, respectively.

- A superior in-sample fit does not by itself lead to the conclusion that a particular forecasting model necessarily produces better forecasts.

Trade-Offs in Model Selection

- Equation (1) compares the in-sample performance of the two models; the same sample $t = 1, \ldots, T$ is used for parameter estimation and model evaluation.
- More highly parameterized models often perform well in comparisons of in-sample fit even when they produce poor forecasts when applied to new data
- Selection of forecasting models involves a bias-variance trade-off
  - Larger (more complex) models exhibits less bias but increased variance due to additional estimation error
  - Smaller models may exclude relevant predictors and leads to greater model misspecification error (bias)
- Example: Risk for subset regressions

Trade-Offs in Model Selection

Example: Risk for subset regressions

- Consider a linear forecasting model $y_{t+1} = \beta' x_t + \varepsilon_{t+1}$, where x_t is a $(K \times 1)$ vector, $\varepsilon_{t+1} \overset{i.i.d.}{\sim} (0, \sigma^2)$ is independent of $Z_t = (x_0', x_1', \ldots, x_T')'$
- Let Z_{kT} be the random variable generated by a subset of $k < K$ predictors and Z_{-kT} generated by the subset of omitted variables so that $Z_{-kT} \cup Z_{kT} = Z_T$.
- Suppose the forecasting model is $f(z_T) = \hat{\beta}_k' x_T$, where $\hat{\beta}_k$ is the OLS estimator for β that includes only $k < K$ regressors; $\hat{\beta}_k$ is a $(K \times 1)$ vector with 0s in place of estimates for the omitted variables.
- Accounting for estimation, the forecast error is

$$
\begin{aligned}
y_{T+1} - \hat{\beta}_k' x_T &= \varepsilon_{T+1} + (\beta - \hat{\beta}_k)' x_T \\
&= \varepsilon_{T+1} + (\beta - E[\hat{\beta}_k | Z_{-kT}])' x_T + (E[\hat{\beta}_k | Z_{-kT}] - \hat{\beta}_k)' x_T
\end{aligned}
$$

Trade-Offs in Model Selection

Example: Risk for subset regressions

- Integrating the squared forecast error only over (y_{T+1}, Z_{kT}) conditional on Z_{-kT} the expected loss becomes

$$
\begin{aligned}
&E\left[\left(\varepsilon_{T+1} + (\beta - E[\hat{\beta}_k|Z_{-kT}])' x_T + (E[\hat{\beta}_k|Z_{-kT}] - \hat{\beta}_k)' x_T\right)^2 \middle| Z_{-kT}\right] \\
&\quad = \sigma^2 + E\left[(\hat{\beta}_k - E[\hat{\beta}_k|Z_{-kT}])' x_T x_T' (\hat{\beta}_k - E[\hat{\beta}_k|Z_{-kT}]) \middle| Z_{-kT}\right] \\
&\qquad + (\beta - E[\hat{\beta}_k|Z_{-kT}])' E[x_T x_T'|Z_{-kT}](\beta - E[\hat{\beta}_k|Z_{-kT}])
\end{aligned}
$$

- The second term is a variance term related to estimation error, which is approximately equal to $\sigma^2 k/T$. The third term is a bias term.
- We refer to the second and third terms collectively as the "estimation effect terms"

Trade-Offs in Model Selection

Example: Risk for subset regressions

- Omit the first term σ^2 and multiply by T, the estimation error term is

$$R_k(\beta) \approx k\sigma^2 + T(\beta - E[\hat{\beta}_k | Z_{-kT}])' E[x_T x_T' | Z_{-kT}](\beta - E[\hat{\beta}_k | Z_{-kT}])$$

- The first term increases as the dimension of the model, k increases.
- The second term is due to model specification error and will generally decrease as k increase.
- The bias-variance trade-off depends on both the true model and on the number of predictors included in the largest model, $K = \dim(x_t)$.
  - When K is small, the estimation error cannot get too large. However with large-dimensional models, the estimation error could potentially become very large.

Trade-Offs in Model Selection

- Sparse models are models for which the number of true nonzero elements of β is small relative to K.
- The misspecification error for sparse models is likely to be small relative to the estimation error associated with large models and so model selection methods that remove many covariates are preferable.
- In situations with many covariates where many elements of β are expected to be nonzero, the misspecification term might be large for all parsimonious models.
  - Model selection methods are useful, but alternative methods such as factor models that consider many covariates might also be useful in this case
- Global search across model specification is employed only for relatively low-dimensional model sets.
  - With K variables in a linear model, we need to consider 2^K possible model specifications.

Sequential Hypothesis Testing

- Sequential hypothesis testing chooses the best submodel from a larger set of available models through a sequence of specification tests such as t-tests or F-tests.
    - General-to-specific (backward stepwise approach): include all potential variables in the initial model and then sequentially remove variables deemed not to be useful
    - Specific-to-general (forward stepwise approach): begin with a small baseline model and then add variables if they appear to improve the prediction model
- Advantage: it is intuitive and computationally simple to implement, particularly if variables are considered one at a time.
- Disadvantage: it does not take a comprehensive search across all possible models, its outcome can be path dependent; there is no guarantee it finds the best possible model within the feasible model set.

Sequential Hypothesis Testing

Example: General-to-specific approach

1. Consider a linear forecasting model with K potential predictor variables. Suppose the OLS is used to estimate the model that includes all predictors:

$$y_{t+1} = \beta_0 + \beta_1 x_{1t} + \beta_2 x_{2t} + \cdots + \beta_K x_{Kt} + \varepsilon_{t+1}$$

2. Suppose the smallest absolute value of the t-statistic falls below some threshold $\underline{t}$, such as $\underline{t} = 2$

$$t_{\min} = \min_{k=1,\dots,K} |t_{\hat{\beta}_k}| < \underline{t}$$

3. The variable associated with $t_{\min}$ is eliminated (assume it is x_{Kt}), then we reestimate the model with the remaining $K-1$ variables.
4. Again, compute the t-statistic for all remaining coefficients, check whether $\min_{k=1,\dots,K-1} |t_{\hat{\beta}_k}| < \underline{t}$, and drop the variable with the smallest t-statistic.
5. This is repeated until the t-statistic for all variables in the model exceed $\underline{t}$ or until some maximum number of iterations is reached.

Sequential Hypothesis Testing

Example: Specific-to-general approach

1. Begin with a simple model that only includes an intercept $y_{t+1} = \beta_0 + \varepsilon_{t+1}$. Each of the K models is next considered separately:

$$y_{t+1} = \beta_0 + \beta_k x_{kt} + \varepsilon_{t+1}, \quad k = 1, \ldots, K$$

2. Suppose the highest absolute value of the t-statistic exceeds some threshold $\bar{t}$

$$t_{\max} = \max_{k=1,\ldots,K} |t_{\hat{\beta}_k}| > \bar{t}$$

   then the variable associated with $t_{\max}$ is included in the model.

3. The remaining $K - 1$ variables are added, one by one, to the model with one variable and intercept. A new variable is included provided that its t-statistic exceeds $\bar{t}$.

4. This is repeated until t-statistics of all of the remaining excluded variables fall below the threshold $\bar{t}$, or until some maximum number of iterations is reached.

Information Criteria

- Information criteria (IC) choose models by trading off model fit against a penalty for model complexity as measured by the number of free parameters that have to be estimated.
- Two commonly used IC: Bayesian Information Criterion (BIC) and Akaike Information Criterion (AIC) has the form

$$IC(k) = -2T^{-1}\ell_T(\hat{\theta}_k) + h(n_k)g(T)$$

  - Each model $M_k \in \mathcal{M}_K$, $k = 1, \ldots, K$, requires estimating n_k number of parameters, θ_k
  - $\ell_T(\theta_k)$ is the log-likelihood function, $\ell_T(\theta_k) = \log L_T(\theta_k)$, where the maximum likelihood estimator $\hat{\theta}_k$ maximizes $\ell_T(\theta)$
  - $h(n_k)$ is an increasing function of n_k
  - $g(T)$ is a decreasing function of the sample size T

Information Criteria

- **Bayesian Information Criterion (BIC)** aka Schwarz Information Criterion (SIC)

$$BIC(k) = -2T^{-1}\ell_T(\hat{\theta}_k) + \frac{n_k \ln(T)}{T}$$

  - The model that minimizes $BIC(k)$ is the model with the highest posterior probability given the data

- **Akaike Information Criterion (AIC)**

$$AIC(k) = -2T^{-1}\ell_T(\hat{\theta}_k) + \frac{2n_k}{T}$$

  - AIC selects the model whose estimated density is closest to the true density.

Information Criteria

Example: AIC and BIC for linear regressions

- Consider a set of linear prediction models that differ in their chosen predictor variables x_{kt}

$$M_k : \quad y_{t+1} = x_{kt}'\beta_k + \varepsilon_{k,t+1}, \quad \varepsilon_{k,t+1} \overset{i.i.d.}{\sim} N(0, \sigma_k^2)$$

- The log-likelihood function is $\ell_T(\theta_k) = \sum_{t=0}^{T-1} \ln p(y_{t+1}|x_{kt}, \theta_k)$, $\theta_k = (\beta_k, \sigma_k^2)$ and $\hat{\sigma}_k^2$ is the MLE estimator of the variance

$$\ell_T(\hat{\theta}_k) = -\frac{T}{2}\ln(2\pi\hat{\sigma}_k^2) - \frac{1}{2\hat{\sigma}_k^2}\sum_{t=1}^{T}\hat{\varepsilon}_{k,t+1}^2 = -\frac{T}{2}\ln(2\pi\hat{\sigma}_k^2) - \frac{T}{2}$$

- $AIC(k) = \ln(\hat{\sigma}_k^2) + \frac{2n_k}{T}$
- $BIC(k) = \ln(\hat{\sigma}_k^2) + \frac{n_k \ln(T)}{T}$

Information Criteria

Other information criteria

- Mallows C_p criterion (Mallows, 1973) is often used to select a submodel from a set of linear regressions

$$C_p(k) = \frac{SSR_k}{\hat{\sigma}^2} - T + 2n_k$$

  - $\hat{\sigma}^2$ is the estimated variance from the "super" model that includes all possible regressors
  - SSR_k is the sum of squared residuals associated with the model with n_k parameters
  - Model selection in linear regressions based on the Mallows C_p is asymptotically equivalent to model selection based on AIC

Information Criteria

Other information criteria

- HQIC (Hannan and Quinn, 1979) is used to select the order of a linear autoregressive (AR) model

$$HQIC(k) = \ln \hat{\sigma}_k^2 + \frac{2k \ln(\ln T)}{T}$$

  - The criterion is based on ensuring consistent estimation of the order of the lag length as well as avoiding overfitting the regression.
  - BIC is also often used to choose the lag length in AR or ARMA models.

Cross-Validation

- In applied statistics and machine learning, the default method for model selection and tuning parameter selection is cross-validation (CV).
- CV avoids overfitting a model by removing the correlation that causes the estimated in-sample loss to be small due to the use of the same observations for both parameter estimation and model evaluation purposes.
- Example: In-sample vs. out-of-sample MSE
  - Forecast y_{T+1} in a sequence of i.i.d. data $\{y_t\}_{t=1}^{T}$ with mean μ and variance σ^2
  - Under MSE loss the best forecast is an estimate of μ, such as $\bar{y}_T = T^{-1} \sum_{t=1}^{T} y_t$
  - In-sample: $E[(y_t - \bar{y}_T)^2] = \sigma^2(1 - T^{-1})$
  - Out-of-sample: $E[(y_{T+1} - \bar{y}_T)^2] = \sigma^2(1 + T^{-1})$

Cross-Validation

Leave-one-out CV

- Consider the linear regression model $y_{t+1} = x_t'\beta + \varepsilon_{t+1}$, $t = 0, \ldots, T-1$; let $\hat{\beta}_{\{t\}}$ be the OLS estimator that uses all observations except t

$$\hat{\beta}_{\{t\}} = \left(\sum_{s \neq t} x_s x_s' \right)^{-1} \sum_{s \neq t} x_s y_{s+1}$$

- The prediction error is $\tilde{e}_{t+1}^2 = y_{t+1} - x_t'\hat{\beta}_{\{t\}}$
- The cross-validated estimator of MSE is

$$CV_1 = \frac{1}{T} \sum_{t=0}^{T-1} \tilde{e}_{t+1}^2$$

- Model selection proceeds by choosing the model that minimizes the criterion CV_1

Cross-Validation

Leave-T_v-out CV

- The leave-T_v-out (v-block) CV approach is developed for independent data.
- The method holds out $T_v = 2v + 1$ observations for model evaluation and the remaining observations are used for parameter estimation.
- Conducted for all possible $\binom{T}{T_v}$ evaluation sets, this results in a very large number of possible evaluation sets and can be computationally costly.

Cross-Validation

v-block CV

- For serially dependent time series the holdout samples are constrained to contiguous observations; there are $T - T_v$ ways to split the sample.
- The v-block CV holds out a block of T_v observations for model evaluation and the remaining are used for parameter estimation.
- The model that minimizes CV_v is chosen

$$CV_v = \frac{1}{T - 2v - 1} \sum_{t=v+1}^{T-v} \left[\frac{1}{2v+1} \sum_{s=t-v}^{s=t+v} (y_{s+1} - f(z_s, \hat{\beta}_{\{t-v:t+v\}}))^2 \right]$$

where $\hat{\beta}_{\{a:b\}}$ is the estimator for β that use all of the sample apart from the observations from a to b with $0 < a < b < T$.

Cross-Validation

hv-block CV

- When the data are serially independent, the usefulness of the CV approach comes from the fact that the parameter estimator is independent of the evaluation sample.
- In practice, predictor variables may include lags of y which would induce correlation between estimation and evaluation samples.
- Shao (1993) shows that v-block cross-validation is inconsistent for linear model selection and will tend to select unnecessarily large models.
- Burman, Chow, and Nolan (1994), Racine (2000) provide methods that remove enough observations so that the correlation between the evaluation and estimation sample is close to 0.

Cross-Validation

hv-block CV

- Racine (2000) proposes to omit h observations on each side of the evaluation sample in order to counter any dependence between the estimation and evaluation samples.
- Similar to v-block CV, a window of v observations before and after time t is used as an evaluation sample of $2v+1$ observations to average of each t.
- In addition, h observations before and after the evaluation sample are eliminated, and the remaining observations are used as the estimation sample.

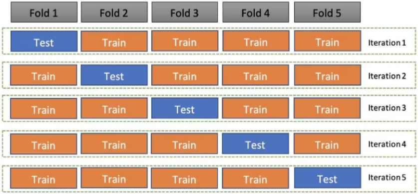
$$CV_{hv} = \frac{1}{T-2v-1-2h} \sum_{t=h+v+1}^{T-v-h} \left[\frac{1}{2v+1} \sum_{s=t-v}^{s=t+v} \left(y_{s+1} - f(z_s, \hat{\beta}_{\{t-h-v:t+h+v\}})\right)^2 \right]$$

- Racine suggested to set $h = 0.25T$ and use an evaluation sample equal to the integer part of $T^{1/2}$

Cross-Validation

K-fold CV

- The most common approach in tuning hyperparameters in machine learning is K-fold cross-validation.



(Image source: sqlrelease.com)

- Split the sample into K groups (or "folds") and treat each group as a hold-out sample.
  - This reduces the number of estimations from $T - T_v$ to K
  - A common choice is $K = 5$ or $K = 10$
  - The classical K-fold CV is inappropriate for time-series data

Cross-Validation

K-fold CV

- Suppose we have a regression model $y = g(x, \theta) + \varepsilon$ with estimator $\hat{\theta}$ and T observations.
  1. Randomly sort the observations.
  2. Split the observations into folds $k = 1, \ldots, K$ of (roughly) equal size $T_k \approx T/K$. Let I_k denote the observations in fold k.
  3. For $k = 1, \ldots, K$
     1. Exclude fold I_k from the dataset. This produces a sample with $T - T_k$ observations.
     2. Calculate the estimator $\hat{\theta}_{(-k)}$ on this sample.
     3. Calculate the prediction errors $\tilde{e}_i = y_i - g(x_i, \hat{\theta}_{(-k)})$ for $i \in I_k$.
     4. Calculate $CV_k = T_k^{-1} \sum_{i \in I_k} \tilde{e}_i^2$.
  4. Calculate $CV = K^{-1} \sum_{k=1}^{K} CV_k$

Cross-Validation

K-fold CV

- K-fold CV can be sensitive to the initial random sorting.
  - This problem can be reduced by repeated CV, which repeats the K-fold CV algorithm M times, each with a different random sorting.
  - The M values of CV are averaged to produce the repeated CV value. As M increases, the randomness due to sorting is eliminated.
- We can think of tuning parameters (selecting hyperparameters) in Machine Learning algorithms as a model selection procedure. The K-fold CV is widely used for Machine Learning.
  - Be aware that the initial random sorting of observations is inappropriate for time-series data.

Lasso Model Selection

- The Lasso (Least Absolute Shrinkage and Selection Operator) was introduced by Tibshirani (1996) as a shrinkage estimator for least squares regression and also turns out to have useful model selection properties.
- The Lasso estimator of β minimizes

$$T^{-1} \sum_{t=0}^{T-1} (y_{t+1} - x_t' \beta)^2 + \lambda \sum_{i=1}^{n_k} |\beta_i| \tag{2}$$

  - $\lambda > 0$ is a hyperparameter that has the effect of shrinking the estimates toward 0.
  - It is conventional to scale all the x variables to have mean 0 and variance 1 before applying Lasso.
- There is no closed form solution for minimizing (2), so computational methods are required.

Lasso Model Selection

- The use of a squared objective function penalized by the absolute value of the parameters results in the property that there are many corner solutions for which $\hat{\beta}_i = 0$
  $\Rightarrow$ Lasso can be used as a model selection tool

- The Lasso minimization problem (2) is equivalent to the dual constrained minimization problem

$$\hat{\beta}_{lasso} = \underset{\sum_{i=1}^{n_k} |\beta_i| \leq \tau}{\arg\min} \; T^{-1} \sum_{t=0}^{T-1} (y_{t+1} - x_t' \beta)^2$$

Lasso Model Selection

- In practical applications, a choice of the regularization parameter λ must be made.
  - Trade-off between complexity and parsimony
- Methods based on theoretical justification provide rate conditions on λ, a function of the sample size T
  - Belloni and Chernozhukov (2011): $\lambda_T = 2c\sigma\sqrt{T}\Phi^{-1}(1-\alpha/2n_k)$
- A common selection method is minimization of K-fold CV; the selected λ aims to select models with good forecast accuracy but not necessarily for other purposes such as accurate inference
  - Chetverikov, Liao, and Chernozhukov (2021): asymptotic consistency of CV selection for Lasso estimation